

GLOSSARY

Fairness (AI Systems)

The principle that AI systems should produce outcomes that are equitable across different demographic groups, minimizing unjustified disparities.

Bias

A systematic and repeatable error in an AI system that results in unfair outcomes for certain groups.

Protected Attributes

Characteristics that should not directly influence decisions due to ethical or legal considerations.

Examples:

- Gender
- Race / Ethnicity
- Nationality
- Age

Proxy Variables

Non-protected features that are strongly correlated with protected attributes and can indirectly introduce bias.

Example:

Travel frequency acting as a proxy for nationality.

Intersectionality

The combined effect of multiple attributes (e.g., gender × nationality) leading to compounded or hidden bias.

Causal Analysis Terms

Causal Fairness Analysis

A structured method to identify *why* fairness disparities exist by analyzing relationships between data, features, and outcomes.

Causal Pathway

The sequence through which bias propagates in a system.

Example:

Nationality → OCR error → Lower confidence → Rejection

Measurement Bias

Bias is introduced due to inaccuracies in data collection or feature measurement.

Example:

OCR systems perform worse on non-Latin scripts.

Root Cause

The underlying mechanism that leads to observed fairness disparities.

Intervention Concepts

Fairness Intervention

An action taken to reduce or eliminate bias in an AI system.

Pre-Processing

Data-level interventions applied before model training to correct bias in datasets.

Examples:

- Rebalancing
- Data cleaning
- Label correction

In-Processing

Model-level interventions applied during training to enforce fairness constraints.

Examples:

- Fairness-aware loss functions
- Proxy feature control

Post-Processing

Decision-level adjustments applied after model predictions to mitigate unfair outcomes.

Examples:

- Threshold adjustment
- Human review for borderline cases

Layered Intervention

Applying multiple interventions across data, model, and decision stages when bias exists at multiple points.

Risk & Governance Concepts

Risk Classification

The process of categorizing AI systems based on potential impact and harm.

High-Risk System

An AI system that significantly affects individuals' rights or opportunities and requires strict governance.

Examples:

- Hiring systems
- Credit scoring
- Immigration decisions

Governance

The framework of policies, roles, and processes that ensure accountability, compliance, and oversight of AI systems.

Escalation

The process of raising issues to higher authority when fairness risks or trade-offs exceed acceptable limits.

Audit Trail

A documented record of decisions, changes, and validations for compliance and traceability.

Validation & Metrics

Validation

The process of verifying that fairness interventions improved outcomes without unacceptable performance loss.

Fairness Metrics

Quantitative measures used to evaluate fairness.

Examples:

- False Positive Rate (FPR)
- False Rejection Rate (FRR)
- Disparate Impact Ratio

Performance Metrics

Measures of model effectiveness.

Examples:

- Accuracy
- Precision
- Recall

Threshold

A decision boundary used to determine outcomes (e.g., approve vs reject).

Drift

A change in data distribution or model behavior over time that can reintroduce bias.

Workflow & System Concepts

Fairness Lifecycle

The continuous process of detecting, analyzing, mitigating, validating, and monitoring bias.

Monitoring

Ongoing evaluation of fairness and performance after deployment.

Feedback Loop

The process of re-entering the workflow when new bias or drift is detected.

Black-Box Model

A system where internal workings are not accessible, limiting intervention options.

Regulatory & Compliance Terms

EU AI Act

A regulatory framework that classifies AI systems based on risk and mandates requirements for high-risk systems.

GDPR Article 22

A regulation granting individuals the right to explanation and human review of automated decisions.

Human Oversight

The ability for humans to intervene in or review AI decisions.

Transparency

The ability to explain how and why decisions are made by an AI system.

Organizational Concepts

RACI Model

A framework defining roles:

- Responsible
- Accountable
- Consulted
- Informed

Fairness Decision Record (FDR)

A documented record of fairness-related decisions, trade-offs, and justifications.

Model Card

A document describing model performance, limitations, and fairness considerations.

This glossary ensures consistent interpretation of fairness concepts across technical, business, and governance stakeholders.